

Javier A. Avecillas-Leon

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EDUCATION

University of Illinois Urbana-Champaign, USA

PhD Candidate in Civil Engineering
Major: Structural Engineering

2020 - present
Current GPA: 4/4

Virginia Tech, USA

MSc in Civil Engineering
Major: Structural Engineering

2017 - 2018
Overall GPA: 4/4

University of Cuenca, Ecuador

BS in Civil Engineering
Major: Civil Engineering

2009 - 2014
Graduation Grade: 90.94/100

ACADEMIC RESEARCH

MSc Thesis

“Topology Optimization of Steel Shear Fuses to Resist Buckling”

Advisor: Matthew Eatherton, PhD

Skills: FEAP FE Models, Abaqus FE Models, Programing, Genetic Algorithms, Technical Writing, Self-learning.

2017 - 2018

BS Thesis

“Implementing Buckling-Restrained Braces as Energy Dissipation Devices”

Advisor: Juan Carrion, PhD

Skills: Designing and manufacturing low-cost small-scale buckling-restrained braces.

2015

RESEARCH PUBLICATIONS

Journals

- **Avecillas-Leon, J.**, de Oliveira Miranda, A. C., Duarte, C. A. (2026). “A Multiscale Framework for Estimating the Fatigue Endurance Limit of Notched Components.” In preparation.
- **Avecillas-Leon, J.**, Bento, M. H. C., Duarte, C. A. (2026). “A three-scale generalized finite element method for the simulation of three-dimensional crack propagation.” Submitted to *Computer Methods in Applied Mechanics and Engineering*
- **Avecillas-Leon, J.**, Singh, I., Duarte, C. A. (2025). “Validation and Improvements of a Generalized/eXtended Finite Element Method for 3-D Fatigue Crack Propagation.” *International Journal of Fracture*, 249(4), 71. [doi:10.1007/s10704-025-00888-6](https://doi.org/10.1007/s10704-025-00888-6)
- Li, H., **Avecillas-Leon, J.**, Shauer, N., Duarte, C. A. (2022). “A non-intrusive iterative generalized finite element method for multiscale coupling of 3-D solid and shell models.” *Computer Methods in Applied Mechanics and Engineering*, 402, 115408. [doi:10.1016/j.cma.2022.115408](https://doi.org/10.1016/j.cma.2022.115408)
- **Avecillas, J. A.**, Eatherton, M. R. (2020). “Controlling out-of-plane buckling in shear-acting structural fuses through topology optimization.” *Journal of Structural Engineering*, 146(7), 04020132. [doi:10.1061/\(ASCE\)ST.1943-541X.0002692](https://doi.org/10.1061/(ASCE)ST.1943-541X.0002692)

Conferences

- **Avecillas, J.**, Eatherton, M. (2019). “Topology optimization of steel shear fuses to resist buckling.” *SSRC Annual Stability Conference, St. Louis, USA*.
- Nguyen, T., **Avecillas, J.**, Eatherton, M. (2019). “Developing design procedures for optimized buckling-resisting structural fuses.” *12th Canadian Conference on Earthquake Engineering, Quebec, Canada*.

PRESENTATIONS

Conference Presentations

- “A Multi-Scale Simulation Framework for Predicting the Fatigue Endurance Limit of Notched Specimens”. Accepted for presentation at the *18th U.S. National Congress on Computational Mechanics*. Chicago, Illinois, July 2025.
- “On the Computational Performance of a Generalized Finite Element Method for Phase-Field Fracture Models”. *17th U.S. National Congress on Computational Mechanics*. Albuquerque, New Mexico, July 2023.
- “Coupling 3-D Solid and Shell Models with a Non-Intrusive Iterative Global-Local Algorithm”. *Engineering Mechanics Institute Conference*. Baltimore, Maryland, May 2022.
- “Iterative Global-Local Algorithm for Coupling 3D Solid and Shell Models”. *16th U.S. National Congress on Computational Mechanics*. Online, July 2021.
- “Topology optimization of steel shear fuses to resist buckling”. *SSRC Annual Stability Conference*. St. Louis, Missouri, April 2019.

FELLOWSHIPS

Karol Fellowship, UIUC

2020

- Department of Civil and Environmental Engineering, University of Illinois at Urbana-Champaign, USA.

Graduate Fellowship, SENESCYT-Ecuador

2015

- National Secretariat for Higher Education, Science, Technology and Innovation - SENESCYT, Ecuador. “Open Summons 2015 - Second Stage”.

AWARDS AND HONORS

Travel Awards

- USNCCM18 Travel Award, U.S. National Congress on Computational Mechanics. 2025
- Conference Travel Award, Department of Civil and Environmental Engineering, UIUC. 2025
- Conference Travel Award, Department of Civil and Environmental Engineering, UIUC. 2022
- Conference Travel Award, Department of Civil and Environmental Engineering, UIUC. 2021

Academic Honors

- Best Academic Record, Colegio Hermano Miguel La Salle (High School), Ecuador. 2004 – 2009
- Best Academic Record, Escuela Superior Luis Cordero (Elementary School), Ecuador. 1997 – 2003

TECHNICAL STRENGTHS

Computer Languages

C++, MATLAB, Python

Technical Software

ABAQUS, LS-DYNA, FEAP, OpenSees, SAP2000, ETABS

Software & Tools

LaTeX, Mathematica, Mathcad, Microsoft Office

RELEVANT COURSES

Core Courses

Fracture Mechanics
Computational Fracture Mechanics
Finite Element Analysis
Nonlinear Finite Elements
Computational Inelasticity
Generalized Finite Element Method

Other Courses

Advanced Mechanics of Materials
Theory of Elasticity
Computational Plates and Shells

WORK EXPERIENCE

University of Illinois Urbana-Champaign, USA

January 2020 - present

Graduate Research Assistant

- Conducting research on multiscale modeling and fatigue life prediction of notched structural components using the Generalized Finite Element Method (GFEM).

University of Illinois Urbana-Champaign, USA

Spring 2026

Teaching Assistant

- Teaching Assistant for CEE 470: Structural Analysis (upper-level undergraduate).
- Supported students with structural analysis theory, grading, and weekly office hours.

University of Illinois Urbana-Champaign, USA

Spring 2025

Teaching Assistant

- Teaching Assistant for CEE 570: Finite Element Methods (graduate level).
- Supported students with FEM theory, assisted with coding, grading, and weekly office hours.

CONSULTBIM, Ecuador

April 2019 - December 2019

Civil Engineer

- Member of the Structural Engineering Department. Structural design and seismic evaluation of existing buildings.

CONSULTBIM, Ecuador

December 2015 - November 2016

Civil Engineer

- Member of the Structural Engineering Department. Structural design and seismic evaluation of existing buildings. Guarantee quality standards and technical contract conditions required for the projects.

COMMUNITY INVOLVEMENT

Ministry of Urban Development and Housing, Ecuador

April 2016

Civil Engineer

- Technical volunteer at the post-earthquake structural evaluation of buildings at Bahia de Caraquez, Ecuador.

San Francisco Rural Community, Ecuador

January 2016

Civil Engineer

- Technical volunteer in the construction of a water irrigation channel to improve the quality of the community's water resources.